

Solving for L_M Step-by-Step

Specific-Factors Model Example

Setup

We have the production functions:

$$Q_M = K^{1/3} L_M^{2/3}, \quad Q_F = T^{1/2} L_F^{1/2}, \quad L_F = 90 - L_M.$$

Given endowments:

$$K = 64 (\Rightarrow K^{1/3} = 4), \quad T = 100 (\Rightarrow T^{1/2} = 10).$$

The equilibrium condition for labor mobility is:

$$P_M \cdot MPL_M = P_F \cdot MPL_F.$$

The marginal products are:

$$MPL_M = \frac{2}{3} K^{1/3} L_M^{-1/3} = \frac{8}{3} L_M^{-1/3}, \quad MPL_F = \frac{1}{2} T^{1/2} L_F^{-1/2} = 5(90 - L_M)^{-1/2}.$$

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Case 1: Initial Prices $P_M = 2$, $P_F = 1$

Step 1: Substitute into the equilibrium condition

$$2 \cdot \frac{8}{3} L_M^{-1/3} = 1 \cdot 5(90 - L_M)^{-1/2},$$

which simplifies to

$$\frac{16}{3} L_M^{-1/3} = 5(90 - L_M)^{-1/2}.$$

Step 2: Take reciprocals to make exponents positive

$$\frac{3}{16} L_M^{1/3} = \frac{1}{5} (90 - L_M)^{1/2}.$$

Step 3: Clear constants and square both sides to eliminate the square root

$$\left(\frac{15}{16}\right)^2 L_M^{2/3} = (90 - L_M),$$

or equivalently,

$$\frac{225}{256}L_M^{2/3} = 90 - L_M.$$

Step 4: Substitute $y = L_M^{1/3}$ so that $L_M = y^3$ and $L_M^{2/3} = y^2$

$$\frac{225}{256}y^2 = 90 - y^3 \Rightarrow y^3 + \frac{225}{256}y^2 - 90 = 0.$$

Step 5: Solve the cubic numerically Testing values:

$$y = 4.20 \Rightarrow f(y) = 4.20^3 + \frac{225}{256}(4.20)^2 - 90 \approx -0.42,$$

$$y = 4.22 \Rightarrow f(y) \approx +0.81.$$

By interpolation, $y \approx 4.2068$ gives $f(y) \approx 0$.

Step 6: Compute L_M

$$L_M = y^3 \approx (4.2068)^3 \approx 74.45.$$

Hence:

$L_M = 74.45, \quad L_F = 90 - 74.45 = 15.55.$
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Case 2: After Price Increase $P_M = 3, P_F = 1$

Step 1: New wage condition

$$3 \cdot \frac{8}{3}L_M^{-1/3} = 5(90 - L_M)^{-1/2} \Rightarrow 8L_M^{-1/3} = 5(90 - L_M)^{-1/2}.$$

Step 2: Take reciprocals and square

$$\left(\frac{5}{8}\right)^2 L_M^{2/3} = 90 - L_M,$$

$$\frac{25}{64}L_M^{2/3} = 90 - L_M.$$

Step 3: Substitute $y = L_M^{1/3}$ again

$$y^3 + \frac{25}{64}y^2 - 90 = 0.$$

Step 4: Solve numerically

$$y = 4.35 \Rightarrow f(y) \approx -0.28,$$

$$y = 4.36 \Rightarrow f(y) \approx +0.33.$$

The root is near $y = 4.3534$.

Step 5: Compute L_M

$$L_M = y^3 \approx (4.3534)^3 \approx 82.59,$$

so

$L_M = 82.59, \quad L_F = 90 - 82.59 = 7.41.$

Summary of Results

Case	L_M	L_F
$P_M = 2$	74.45	15.55
$P_M = 3$	82.59	7.41

As P_M rises, labor moves into manufacturing, confirming the positive link between the relative price of M and L_M predicted by the specific-factors model.